MIL-M-46728C
25 JULY 1983
Superseding
MIL-M-46728B
28 July 1979

MILITARY SPECIFICATION

MIRROR ASSEMBLY, REARVIEW: AUTOMOTIVE, EXTERIOR MOUNTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers exterior-mounted, automotive rearview mirror assemblies.

1.2 Classification. Rearview mirror assemblies shall be of the following types (see 6.2):

Type I	- Stud, ball and socket mount.
Type II	- Top and bottom stud mount.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

DD-G-451	- Glass, Flat and Corrugated, for Glazing, Mirrors and Other Uses.
PPP-T-60	- Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon.
TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

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MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-C-81562	- Coatings, Cadmium, Tin-cadmium, and Zinc (Mechanically Deposited).

STANDARDS
FEDERAL

FED-STD-H28/2	- Screw Thread Standards for Federal Services.
FED-STD-151	- Metals: Test Methods.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MS53015	- Mirror Assembly, Rearview.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Safety Standard

FMVSS No. 111 - Rearview Mirrors.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, DC 20591.)

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issues listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A108	- Steel Bars, Carbon, Cold-Finished, (Standard Quality).
ASTM B380	- Corrosion Testing of Decorative Chromium Electroplating by the Corrodkote Procedure.
ASTM B633	- Electrodeposited Coatings of Zinc on Iron and Steel.

(Copies of the above publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
SAE Standards and Recommended Practices

J126	- Selecting and Specifying Hot and Cold Steel Sheet and Strip.
J964	- Test Procedures for Determining Reflectivity of Rearview Mirrors.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Avenue, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article inspection. A first article of the rearview mirror assemblies to be furnished under this specification shall be produced prior to the manufacture of item in production quantity. The first article shall be submitted to the Government for first article inspection to determine conformance to the quality assurance provisions of this specification (see 4.3). The first article submitted by a contractor shall be fully representative of the rearview mirror assemblies to be supplied from production facilities and tooling (see 6.3).

3.2 Materials. Materials shall be as specified herein and on applicable standards and specifications (see 6.4).

3.2.1 Mirror glass. Mirror glass shall conform to Type I, float glass, quality q1 - double strength thickness (0.115 minimum - 0.134 inch maximum) of DD-G-451.

3.2.1.1 Mirror coating. The reflecting surface of the mirror coating shall be a silvered coating, uniform in color, and free from stains, scratches, blisters, and misses.

3.2.1.1.1 Chromium. Vacuum metalized or sputtered chromium may be used in place of base silvering operation. The reflective surface is produced by the vacuum deposition of a chromium or chromium alloy film (ASTM B380).

3.2.2 Mirror case. Mirror case shall be of low-carbon steel, 0.024 inch in thickness, cold-rolled commercial quality (CQ), temper 2 to 5 in accordance with QQ-S-698 and SAE J126.

3.2.2.1 Mirror case preparation. Mirror case material shall be cleaned according to the applicable method of TT-C-490 and pretreated in accordance with Type III of TT-C-490.

3.2.2.2 Painting. Exterior finish of mirror case shall be enamel, lusterless, and forest green in accordance with MIL-E-52798 or MIL-E-52835.

3.2.3 Mirror mounts.

3.2.3.1 Type I. Stud, ball and socket of Type I mirror assemblies shall be of carbon steel conforming to compositions C1020 to C1035 of ASTM A 108 and shall be zinc plated in accordance with ASTM B 633, or Type II, Class 3 of MIL-C-81562.

3.2.3.2 Type II. Mounting studs of Type II mirror assemblies shall be of carbon steel conforming to compositions C1020 to C1035 of ASTM A 108 and shall be zinc plated in accordance with ASTM B 633 or Type II, Class 3 of MIL-C-81562.

3.3 Design and construction. All parts shall conform to and be functionally and dimensionally interchangeable with those covered by MS53015.

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3.3.1 Threaded parts. All screw threads shall conform to screw thread standards in accordance with FED-STD-H28/2 (see 2.1).

3.4 Performance.

3.4.1 Field of view. Mirror shall provide a field of view to the rear as specified in Federal Motor Vehicle Safety Standard (FMVSS) No. 111 (see 2.1.2). FMVSS No. 111 applies both to plane (flat) rearview mirrors and to convex (glass surface curved outward) rearview mirrors. In the event of conflict between the foregoing standard and this specification, requirements of this specification shall prevail.

3.4.2 Specular reflectivity. With an angle of incidence of 45 degrees, total reflective light in the visible range shall exceed 50 percent (follow SAE J964 for test procedures).

3.4.3 Range of convexities. Convex mirror shall have an average radius of curvature not less than 20 inches and not more than 60 inches.

3.4.4 Vibration resistance. When subjected to specified vibration, mirror assembly (MS53015-1) shall evidence no disorientation of parts or visible mechanical damage.

3.4.5 Temperature stability. When subjected to specified temperatures, mirrors shall evidence no discoloration, silvering deterioration, or reduction in reflectivity.

3.4.6 Abrasion resistance. When subjected to abrasion and cleaning as specified, the reflectivity of the mirror shall not be reduced by more than 2 percent.

3.4.7 Corrosion resistance. When subjected to corrosive solution as specified, mirror shall evidence no discoloration, silvering deterioration, or any visually apparent reduction in reflectivity.

3.4.8 Coating adhesion. When subjected to coating adhesion testing as specified, there shall be no separation of reflective coating from the mirror.

3.5 Marking. Marking shall be in accordance with MIL-STD-130.

3.6 Workmanship. The workmanship on the mirror assembly shall produce a quality product free from defects affecting its serviceability and appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and materials inspection. The contractor is responsible for assuring that components and materials are examined and tested in accordance with the requirements of the referenced documents to the extent specified herein.

4.1.2 Contractor quality assurance system. The contractor shall provide and maintain an effective inspection and quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall not be restricted to the method of inspection listed providing that an equivalent control is included in the approved quality assurance procedure. The contractor shall notify the Government of and obtain approval for any change to the submitted procedure that might affect the degree of control required by this specification or other applicable documents referenced herein.

4.1.3 Government verification. All quality assurance operations performed by the contractor shall be subject to Government verification at unscheduled intervals. Verification shall consist of surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied and Government product inspection to measure quality of product offered for acceptance. Deviation from prescribed or agreed-upon procedures or instances of poor practices which might have an adverse effect upon the quality of the product, shall immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.2 Classification of inspections. Classification of inspections shall be as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Control tests (see 4.4.5).

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4.3 First article inspection. A first article sample consisting of four mirror assemblies of each type to be furnished, shall be submitted for first article inspection (see 6.3). The samples shall be representative of the units proposed to be furnished under the contract. First article inspection shall be conducted by the Government at a place designated by the acquisition activity and shall consist of examination for the defects specified in 4.4.2.2 and testing as specified in table I.

4.3.1 Failure. Failure of a first article sample to pass any examination or test specified herein shall be cause for Government refusal to conduct additional testing until faults revealed by the test have been corrected.

TABLE I. Test sequence.

Test	Requirement	First article	Quality conformance inspection	Control
Specular reflectivity	3.4.2	4.5.1	4.5.1	4.5.1
Range of convexities	3.4.3	4.5.2	4.5.2	4.5.2
Vibration resistance	3.4.4	4.5.3	4.5.3	4.5.3
Temperature stability	3.4.5	4.5.4	4.5.4	
Abrasion resistance	3.4.6	4.5.5	4.5.5	
Corrosion resistance	3.4.7	4.5.6		4.5.6
Coating adhesion	3.4.8	4.5.7	4.5.7	4.5.7

4.4 Quality conformance inspection.

4.4.1 Sampling.

4.4.1.1 Lot formation. Unless otherwise specified (see 6.2), a lot shall consist of all mirror assemblies of one part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.4.1.3 Sampling for acceptance testing. Sampling for acceptance testing shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.4.2 Quality conformance examination.

4.4.2.1 Acceptable quality level. Each sample selected in accordance with 4.4.1.2 shall be examined for conformance to the following acceptable quality levels (AQLs) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

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4.4.2.2 Classification of defects. For examination purposes, defects shall be classified as shown in table II:

TABLE II. Classification of defects.

Categories	Defects	Method of Inspection
Major		
101	Dimensions affecting interchangeability not within tolerance (see 3.3).	Scale and SIE 1/
102	Field of view, improper (see 3.4.1).	Visual
Minor		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	Scale and SIE 1/
202	Coating imperfections (see 3.2.1.1).	Visual
203	Incorrect finish (see 3.2.2.2).	Visual
204	Screw thread nonconformity (see 3.3.1).	SIE 1/
205	Incorrect or illegible marking (see 3.5).	Visual
206	Poor workmanship (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3 Classification of tests. Tests shall be classified as follows:

a. Acceptance tests (see 4.4.4).
b. Control tests (see 4.4.5).

4.4.4 Acceptance tests. Each sample selected in accordance with 4.4.1.3 shall be examined for the defects as specified in table II and tested as specified in table I using an AQL on the basis of percent defective.

4.4.4.1 Test failure. If the sample fails to pass any acceptance tests specified herein, the Government shall stop acceptance of subsequent samples until the contractor provides evidence of correction.

4.4.5 Control tests. Control tests shall be selected at the rate of two mirror assemblies per month or two of each 500 produced, whichever occurs first. Not more than four mirror assemblies shall be selected in any single month. The first sample shall be taken at or near the start of production and shall be subjected to the applicable tests as specified in table I.

4.4.5.1 Test failure. Failure of a control test sample to pass any specified examination or test may be cause for the Government to refuse subsequent lots until it has been proved to the satisfaction of the Government that faults revealed by the tests have been corrected.

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4.5 Methods of inspection.

4.5.1 Specular reflectivity test. To determine conformance to 3.4.2, specular reflectivity shall be tested in accordance with SAE J964. The Hunter Laboratory Colorimeter (D25-2 or D25-9) or Gardner Laboratory Haze Meter are instruments utilized for testing reflectivity.

4.5.2 Range of convexities test. To determine conformance to 3.4.3, range of convexities shall be measured with a spherometer.

4.5.3 Vibration resistance test. To determine conformance to 3.4.4 using its normal point of installation, the mirror assembly (MS53015-1 only) shall be mounted on a vibration machine. The mirror assembly shall be mounted on the vibration brackets without application of tension or compression from the vibration brackets to the mirror assembly. Mirror assembly shall be vibrated for 1 hour in each of its three axes. Vibration shall be in a periodic cycle from 10 to 55 and back to 10 cycles per second (cps) through an amplitude of 0.030 inch (total excursion 0.060 inch). At completion of test, the mirror assembly shall be examined for defects.

4.5.4 Temperature stability test. To determine conformance to 3.4.5, a disassembled mirror shall be placed in a cold chamber and the temperature reduced to $-65^{\circ}F \pm 5^{\circ}F$ for 2 hours. The mirror shall then be removed from the cold chamber and returned to room temperature. It shall then be placed in an oven with ambient temperature of $200^{\circ}F$ for 2 hours. Possible reduction in reflectivity shall be measured visually comparing the reflected shade of a sheet of white paper with actual shade of the paper.

4.5.5 Abrasion resistance test. To determine conformance to 3.4.6, the reflecting surface shall be cleaned with a soft, unbleached Canton flannel cloth and dusted on the nap side with dry soil passed through a 200-mesh sieve. The reflecting surface shall be rubbed with the cloth for 30 seconds using a rotary motion. The foregoing cleaning cycle shall be repeated 30 times. Mirror shall be examined for any visible damage due to the above abrading and shall then be tested as specified in 4.5.1.

4.5.6 Corrosion resistance test. To determine conformance to 3.4.7, a disassembled mirror shall be subjected to the corrosion test specified in Method 811.1 of FED-STD-151. Salt solution shall be 20 percent by weight. Length of exposure shall be 50 hours $\pm$ 1 hour. At completion of test and prior to mirror being inspected, residue shall be removed by cleaning with a soft, unbleached Canton flannel cloth and shall be examined for defects.

4.5.7 Coating adhesion test. To determine conformance to 3.4.8, a strip of pressure-sensitive tape, 1 inch wide, conforming to PPP-T-60, shall be securely applied to the coated side and over the edges of mirror glass in an ambient air temperature of $68^{\circ}F \pm 2^{\circ}F$. The tape shall be stripped off in one slow, steady motion. The coated surface shall then be examined for defects.

4.6 Inspection preparation for delivery. Preservation, packaging, packing, and marking of mirror assemblies shall be examined to determine conformance to the requirements of Section 5 of this specification.

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet as specified by the acquisition activity (see 6.2).

6. NOTES

6.1 Intended use. Mirror assemblies covered by this specification are intended for use on automotive equipment for rear vision purposes.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type of mirror assembly to be furnished (see 1.2).
c. Military part number (see 3.3).
d. Marking required (see 3.5).
e. Place where first article samples shall be sent (see 4.3).
f. Size of lot if different from that specified (see 4.4.1.1).
g. Selection of applicable packaging standard or packaging data sheet (see 5.1).

6.3 First article. When a first article is required, it shall be tested and approved under appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and approval of first article (see 3.1).

6.4 Recycled materials. The use of recycled materials which meet the requirements of applicable material specifications without jeopardizing intended use of the item shall be encouraged.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of changes.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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